Martin Cooler-Stith

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EDUCATION

Bachelor of Electrical Engineering - Expected May 2028
College of Science and Engineering, University of Minnesota Twin Cities
GPA: 3.5/4.0

Relevant Coursework Completed: Signals, Circuits & Electronics; Analog & Digital Electronics; Introduction to Digital System Design; Introduction to Microcontrollers

Current Coursework: Signals & Systems; Circuits & Electronics Lab I; Analog Electronics; Semiconductor Devices

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Git & GitHub
Embedded Systems: PIC24 firmware development, ultrasonic sensors, PWM motor control
CAD and Design: Fusion 360, Onshape, Blender
Other: Microsoft Office, soldering and debugging circuits

PROJECT EXPERIENCE

Research Assistant; Reinforcement Learning for Quadcopters | Prof. Nicola Elia | May 2026-Present
- Built containerized workflows for training quadcopters to fly using reinforcement learning.
- Integrated Google Cloud services to support cloud-based training.
- Designed and constructed a low-cost tachometer for measuring the angular velocity of drone propellers in the lab.
- Fabricated a flexible drone mount for in-place quadcopter flight testing using both 3D-printed and laser-cut components.

Maze-Solving Robot | EE 2361: Introduction to Microcontrollers | 2026
- Developed PIC24 firmware for a maze-solving robot.
- Used readings from three ultrasonic distance sensors to navigate mazes.
- Troubleshot hardware issues in the lab with an oscilloscope.
- Controlled two brushed DC motors using PWM for speed-controlled motion.

Mesh Network | 2024
- Developed and implemented a network of three ESP32 nodes using ESP-NOW protocol.
- Performed range testing on nodes to determine signal area.

LEADERSHIP EXPERIENCE

Community Manager | Beacon Social Club | Nov. 2025-Present
- Resolved a conflict between community members by talking with them about the situation and helping them reach a peaceful resolution.

CAD Lead | FIRST Tech Challenge Team #14575, Capital Bots | 2021-2023
- Led the team's Fusion 360 mechanical design.
- Mentored new members as they learned robotics.

VOLUNTEER EXPERIENCE

Volunteer Intern | KID Museum, Bethesda, MD | Sep. 2022-May 2023
- Assisted children with Snap Circuits projects, taught technical skills, and encouraged their interest in science & engineering.
- Helped children learn to code with Scratch and supported their creativity.

Volunteer Intern | Greenbelt Access Television, Greenbelt, MD | Sep. 2021-May 2022
- Set up lights and props for local film projects and operated a camera.
- Collaborated with the director.

Volunteer Intern | The SPACE Free Art for All, Greenbelt, MD | Sep. 2020-May 2021
- Tutored students in Scratch and produced recorded Python programming lessons with custom animations.
- Welcomed visitors to the makerspace.